



Financial Literacy in Italy: A Comprehensive Analysis

Pasquale Gravante

July 2024

Contents

1	Introduction	3
1.1	Objective of the study	3
1.2	Data description	3
2	Exploratory Data Analysis	5
2.1	Descriptive Statistics	5
2.2	Financial Literacy Scores Across Demographics	6
3	Determinants of Financial Literacy Levels	7
3.1	Research Question	7
3.2	Methodology	7
3.3	Results	8
3.4	Interpretation	8
4	Regional and Gender-based Disparities	8
4.1	Methodology	8
4.2	Results	9
4.2.1	Regional Disparities Model	9

4.2.2	Gender and Socioeconomic Interaction Model	9
4.3	Interpretation	9
5	Patterns in Financial Behaviors and Attitudes	9
5.1	Research Question	9
5.2	Methodology	10
5.3	Results	10
5.3.1	Feature Importance	10
5.3.2	Association Rules	10
5.3.3	Explanation	10
5.3.4	Practical Implications	10
6	Digital Finance	11
6.1	Data Integration	11
6.2	Research Question	11
6.3	Methodology	11
6.4	Results	11
6.4.1	Multiple Linear Regression	11
6.4.2	ANOVA Results	12
6.5	Implications	13
6.6	Risk Perception	13
6.6.1	Research Question	13
6.6.2	Methodology	13
6.6.3	Results and Interpretation	14
7	Conclusions and Recommendations	16
7.1	Future research	16

Abstract

This study examines financial literacy among Italian adults using data from a Banca d'Italia survey conducted as part of an OECD initiative. Employing various statistical techniques, including ordinal and linear regression with interaction terms, the research investigates the determinants of financial literacy and explores regional and gender disparities. Key findings reveal significant influences of education, employment status, age, and gender on financial literacy levels, with notable regional variations. The study uncovers important interactions, particularly between gender and socioeconomic factors, in shaping financial literacy. These insights provide valuable guidance for policymakers and educators in designing targeted financial education programs to enhance financial literacy across diverse population segments in Italy.

1 Introduction

Financial literacy has emerged as a critical skill in today's complex economic landscape. As individuals face increasingly sophisticated financial products and bear greater responsibility for their financial decisions, understanding the factors that contribute to financial literacy becomes crucial. This study focuses on financial literacy in Italy, aiming to provide a comprehensive analysis of its determinants and variations across different demographic groups.

1.1 Objective of the study

The main objective of this study is to translate research findings into actionable insights for policymakers and educators, facilitating the development of tailored financial education initiatives and informed policy decisions aimed at improving financial literacy across diverse segments of the Italian population.

To do so, I decided to:

- Assess the current levels of financial literacy across various demographic groups in Italy.
- Identify and analyze key sociodemographic and economic factors influencing financial literacy.
- Examine regional disparities in financial literacy and explore potential underlying causes.
- Investigate gender differences in financial literacy and how they interact with other socioeconomic factors.
- Integrating more recent data to also study literacy about digital finance matters.

1.2 Data description

This study utilizes data from a comprehensive survey conducted by Banca d'Italia as part of a broader OECD initiative to create an internationally comparable dataset on financial literacy and inclusion.

Key features of the dataset include:

- **Sample Size:** Approximately 2,500 respondents
- **Data Collection Methods:** Face-to-face interviews and tablet-based responses
- **Key Variables:** Answers to questions regarding:
 - *Demographic Information:* Age, gender, educational qualification, employment status, geographic location
 - *Financial Knowledge:* Scores based on responses to financial concept questions
 - *Financial Behaviors:* Data on financial practices and decision-making
 - *Financial Attitudes:* Information on attitudes towards financial management
 - *Financial Score:* Aggregated index which summarises the answers to all the different kind of questions.

The dataset provides a rich source of information, allowing for a nuanced analysis of financial literacy across various segments of the Italian population. It includes detailed questions on financial knowledge, behaviors, and attitudes, enabling the creation of comprehensive financial literacy scores. The inclusion of a wide range of sociodemographic variables permits an in-depth examination of the factors influencing financial literacy and the exploration of interactions between these factors.

2 Exploratory Data Analysis

This section presents a comprehensive exploratory data analysis of our financial literacy dataset, providing insights into the distribution of key variables and their relationships with financial literacy scores.

2.1 Descriptive Statistics

We begin by examining the distribution of our sample across various demographic categories.

Table 1: Distribution of Age Groups

Age Group	Count
18-29	540
30-44	667
45-59	517
60+	154
under 18	477

Table 2: Distribution of Education Levels

Education Level	Count
No education	26
Primary	884
Secondary	916
University	529

Table 3: Distribution of Regions

Region	Count
Centre	454
North	1109
South & Islands	792

Table 4: Distribution of Employment Status and Gender

Category	Count
Employed	1130
Not in labor force	493
Retired	571
Student	161
Female	1198
Male	1157

Tables 1, 2, 3, and 4 show the distribution of our sample across various demographic categories. We observe a good representation across different age groups, with a slight underrepresentation of individuals over 60. Education levels are skewed towards secondary education and above. The sample has a higher representation from the North. The majority of respondents are employed, with a good representation of retired individuals and those not in the labor force. There's a nearly even distribution between male and female respondents.

2.2 Financial Literacy Scores Across Demographics

We now examine how financial literacy scores vary across different demographic groups through visualizations.

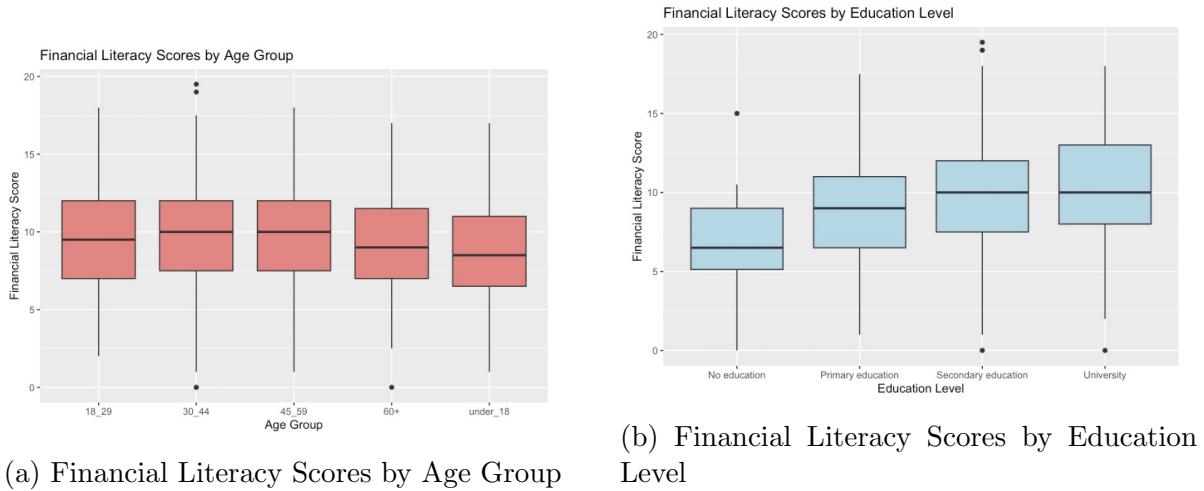


Figure 1: Financial Literacy Scores by Age and Education

- **Age (Fig. 1a):** Middle-aged groups (30-44 and 45-59) show higher median scores, while the under-18 group has the lowest median score. This suggests that financial literacy tends to improve with age and life experience, possibly due to increased exposure to financial decision-making.
- **Education (Fig. 1b):** There's a clear positive correlation between education level and financial literacy scores. University graduates demonstrate the highest median score and the least variability, indicating that higher education may provide more opportunities for financial learning and application.



Figure 2: Financial Literacy Scores by Employment and Sex

- **Employment (Fig. 2a):** Employed individuals show the highest median score, followed closely by retirees. Students and those not in the labor force have lower

scores, highlighting the potential impact of practical financial experience gained through employment.

- **Sex (Fig. 2b):** Males show a slightly higher median score than females, but the difference appears minimal. This suggests a small gender gap in financial literacy, which could be due to various socio-economic factors and deserves further investigation.

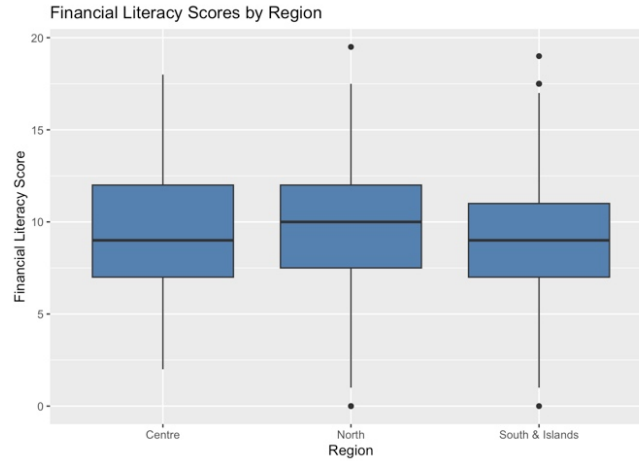


Figure 3: Financial Literacy Scores by Region

- **Region:** The North shows a marginally higher median score, but regional differences appear less pronounced compared to other demographic factors. This could indicate that regional economic disparities have a limited impact on financial literacy, or that other factors are more influential in determining financial literacy levels across Italy.

3 Determinants of Financial Literacy Levels

3.1 Research Question

This question aims to identify and analyze the primary determinants of financial literacy, including factors such as age, education, income, employment status, and region. It also seeks to understand how these factors may interact, providing insights into the complex landscape of financial literacy in Italy.

3.2 Methodology

To have results that could be generalized, the financial literacy score was divided into ordered levels (Very Low, Low, Average, High, Very High). For this reason, having an ordered dependent variable, to analyze the factors influencing financial literacy levels, an ordinal regression model was employed.

3.3 Results

The ordinal regression model yielded the following key results:

Table 5: Ordinal Regression Results for Financial Literacy Determinants

Variable	Coefficient	Std. Error	p-value
Age 30-44	0.30555	0.10864	0.00492
Age 45-59	0.40819	0.16668	0.01433
Education.L	1.45323	0.26326	3.39e-08
Region - North	0.23977	0.10542	0.02295
Employment - Not in labor force	-0.44756	0.10990	4.65e-05
Employment - Student	-0.52058	0.18729	0.00544

3.4 Interpretation

The results indicate that:

- Age significantly influences financial literacy, with middle-aged groups showing higher literacy levels compared to the reference group (18-29).
- Education has a strong positive association with financial literacy, meaning that the more educated, the more financially literated.
- Regional differences exist, with people from the northern regions showing higher financial literacy levels.
- Employment status is significant, with those not in the labor force or students showing lower financial literacy compared to employed individuals.

These findings suggest that targeted financial education programs should focus on younger individuals, those with lower education levels, and those not in the workforce. Regional disparities also indicate a need for tailored approaches in different parts of Italy.

4 Regional and Gender-based Disparities

In this section the possible existence of region-based and gender-based disparities regarding financial literacy is explored.

4.1 Methodology

To further explore regional and gender-based disparities, employed two linear regression models were employed:

1. A regional disparities model examining interactions between region and education/employment.
2. A gender and socioeconomic interaction model focusing on how gender interacts with education, employment, and age.

4.2 Results

4.2.1 Regional Disparities Model

Key results from the regional disparities model (Adjusted R-squared: 0.08229):

Table 6: Significant Coefficients in Regional Disparities Model

Variable	Coefficient	Std. Error	p-value
employmentNot in labor force	-0.92486	0.40133	0.02128
ageunder_18	-0.58290	0.22004	0.00813

Notably, interaction terms between region and education/employment were not statistically significant.

4.2.2 Gender and Socioeconomic Interaction Model

Key results from the gender interaction model (Adjusted R-squared: 0.08976):

Table 7: Significant Coefficients in Gender Interaction Model

Variable	Coefficient	Std. Error	p-value
Education.L	1.64731	0.63008	0.00899
Employment - Not in labor force	-0.51716	0.22760	0.02317
Region - North	0.37956	0.17208	0.02750
Sex = Male & Education	2.36899	0.86444	0.00618
Sex = Male & Employment = Not in labor force	-0.87067	0.38804	0.02494

4.3 Interpretation

- The lack of significant regional interactions implies that the effects of education and employment on financial literacy are consistent across regions. The regional disparities model suggests that employment status and age are significant factors, with those not in the labor force and younger individuals showing lower financial literacy.
- The gender interaction model reveals that the impact of education on financial literacy is stronger for males, while being out of the labor force has a more negative impact on financial literacy for males than for females. These results highlight the need for gender-specific approaches in financial education initiatives.

5 Patterns in Financial Behaviors and Attitudes

5.1 Research Question

What are the underlying patterns and relationships between various financial behaviors and attitudes among Italian adults, and how can these patterns inform targeted financial education initiatives?

5.2 Methodology

To identify important factors and uncover patterns in financial behaviors and attitudes, random forest for feature selection followed by association rule mining were used.

5.3 Results

5.3.1 Feature Importance

The top features influencing financial literacy included various financial attitudes (e.g., qf10_2, qf10_4, qf10_6) and specific financial knowledge questions (e.g., qk5, qk7_3).

5.3.2 Association Rules

Two key rules emerged:

1. $\{qk7_3=1, qk6=1, qk3=3\} \longrightarrow \{qk5=102\}$ (Confidence: 0.91, Lift: 1.66)
2. $\{qf10_2=5, qf10_6=1\} \longrightarrow \{qf10_4=1\}$ (Confidence: 0.87, Lift: 1.57)

5.3.3 Explanation

The feature importance results suggest that both attitudes and specific knowledge play crucial roles in overall financial literacy. More specifically:

- Rule 1: Individuals who correctly answer three specific financial knowledge questions are highly likely (91% probability) to understand the concept of interests correctly.
- Rule 2: Individuals who strongly disagree with "living only for today" and strongly agree with "keeping a close watch on their financial affairs" are very likely (87% probability) to be always on time with paying their bills/other expenses.

5.3.4 Practical Implications

First rule:

- This rule suggests a strong interrelation between certain areas of financial knowledge.
- In practical terms, if an individual understands some concepts they are very likely to know, or to be able to understand, new ones.
- For financial educators, this implies that these concepts could be taught together as a cohesive module, as mastery in one area strongly predicts mastery in the others.

Second rule:

- This rule demonstrate a strong connection between long-term financial attitudes and responsible financial behavior.
- Financial education programs could focus on developing these attitudes (future-oriented thinking and financial vigilance) as a way to indirectly improve concrete financial behaviors.

- For policy makers, this implies that campaigns promoting long-term financial planning and regular financial check-ups could have positive effects on reducing late payments and associated financial stress.

6 Digital Finance

6.1 Data Integration

In this section, new data is introduced from a recent survey conducted in 2022, which includes information about digital financial knowledge and income levels. This dataset allows us to explore more current trends in financial literacy, with a particular focus on digital aspects and their relationship with socioeconomic factors.

The new dataset includes:

- Digital financial literacy scores, calculated based on knowledge, behavior, and attitude answers
- Income level informations
- Measures of digital risk perception

6.2 Research Question

This data enables us to address the following research question:

- How do digital financial literacy levels vary across different income groups, and what other factors influence digital financial literacy?

6.3 Methodology

Two main analytical approaches were employed:

1. Multiple linear regression to identify determinants of digital financial literacy, with a focus on income.
2. ANOVA to assess differences in digital literacy across categorical variables, including income groups.

6.4 Results

6.4.1 Multiple Linear Regression

A multiple linear regression was conducted with digital literacy score as the dependent variable and various demographic factors, including income, as predictors.

Table 8: Multiple Linear Regression Results for Digital Financial Literacy

Variable	Coefficient	Std. Error	p-value
Intercept	48.05529	2.02039	2e-16
Age	-0.06964	0.03559	0.05046
Education.L	10.80806	1.94519	2.98e-08
Income.L	2.50703	0.88076	0.00445
Male	1.83799	0.75540	0.01502
Not in labor force	-6.36575	1.33614	1.98e-06
Student	-9.63871	2.30136	2.89e-05

The multiple linear regression results reveal several key findings:

- Income shows a significant positive relationship with digital financial literacy. This suggests that individuals with higher income tend to have higher digital financial literacy scores.
- Education emerges as the strongest predictor of digital financial literacy, suggesting that higher education levels are associated with better digital financial skills, even when controlling for income.
- Employment status plays a crucial role, with those not in the labor force showing significantly lower digital financial literacy scores compared to employed individuals.
- There's a slight gender effect, with males showing marginally higher digital literacy scores.
- Age has a small negative association with digital literacy, though it's only marginally significant.

These findings suggest that while income is an important factor in digital financial literacy, it's part of a complex interplay of socioeconomic factors. Education, in particular, appears to be a critical determinant of digital financial skills.

6.4.2 ANOVA Results

ANOVA tests were conducted to examine differences in digital literacy across categorical variables, including income groups.

Table 9: ANOVA Results for Digital Financial Literacy

Variable	F value	p-value	Significance
Income	22.11	2.92e-10	***
Education	54.12	2e-16	***
Region	8.674	0.000175	***
Employment	39.84	2e-16	***

The ANOVA results provide additional insights into group differences in digital financial literacy:

- **Income:** The significant F-value confirms substantial differences in digital financial literacy across income groups. This aligns with the linear regression results and emphasizes the strong association between income levels and digital financial skills.
- **Education:** The highest F-value among all variables underscores the critical role of education in determining digital financial literacy levels. This further supports the regression finding that education is the strongest predictor.
- **Employment:** The high F-value indicates significant variations in digital financial literacy based on employment status, consistent with the regression results showing lower scores for non-employed groups.
- **Region:** While significant, the regional differences appear less pronounced compared to other factors. This suggests that while geographic location plays a role, it may be less influential than socioeconomic factors like education and income.

These ANOVA results complement the regression analysis by confirming significant group differences across key categorical variables, with education and work situation showing particularly strong effects.

6.5 Implications

The findings on digital financial literacy, income, and related factors have several key implications:

- **Educational Interventions:** Integrate digital financial literacy into school curricula and develop targeted adult education programs, especially for lower education and income groups.
- **Policy and Regional Focus:** Develop region-specific policies and income-based support systems to address local challenges and socioeconomic disparities in digital financial literacy.
- **Age and Gender Considerations:** Create targeted programs for older adults to improve their digital financial skills and ensure gender-inclusive design in financial education materials and outreach programs.

6.6 Risk Perception

6.6.1 Research Question

- What is the relationship between digital financial literacy and other factors with risk perception in digital financial contexts?

6.6.2 Methodology

Ordinal logistic regression was employed with the variable representing the question regarding risk perception as dependent (ordered from 1 to 5), to analyze how digital financial literacy scores and other factors relate to levels of perceived digital financial risk.

6.6.3 Results and Interpretation

Table 10: Ordinal Logistic Regression Results for Digital Risk Perception

Variable	Coefficient	Std. Error	p-value
Digital Literacy Score	-0.03374	0.00199	2e-16
Education.C	-0.34705	0.11456	0.00247
Income.L	0.28444	0.08484	0.00080
Age	-0.01473	0.00252	4.86e-09
Male	0.15482	0.07324	0.03447

Regression Results: The regression results show that all included variables have a statistically significant relationship with risk perception:

- Digital Literacy Score has a strong negative association with risk perception.
- Education (cubic term) is negatively associated with risk perception.
- Income shows a positive linear relationship with risk perception.
- Age has a negative relationship with risk perception.
- Being male is associated with slightly higher risk perception.

Table 11: Odds Ratios with 95% Confidence Intervals

Variable	OR	2.5%	97.5%
Digital Literacy Score	0.967	0.963	0.971
Education.L	0.932	0.543	1.607
Education.Q	0.913	0.606	1.372
Education.C	0.707	0.565	0.885
Income.L	1.329	1.126	1.570
Income.Q	0.850	0.750	0.964
Age	0.985	0.981	0.990
Male	1.167	1.011	1.348
Region North	1.048	0.866	1.268
Region South & Islands	0.991	0.810	1.211

Log Odds Interpretation The coefficients in the ordinal logistic regression represent the change in log odds of being in a higher risk perception category for a one-unit increase in the predictor variable. Table 11 shows the odds ratios and their confidence intervals:

- Digital Literacy Score: For each unit increase, the odds of being in a higher risk category decrease by 3.3%
- Education: The cubic term shows that higher levels of education are associated with lower perceived risk.

- Income: The linear term indicates that higher income is associated with higher perceived risk. The odds of being in a higher risk category increase by 32.9% for each unit increase in the income scale.
- Age: For each year increase in age, the odds of being in a higher risk category decrease by 1.5%
- Gender: Being male increases the odds of being in a higher risk category by 16.7%
- Region: The effects of region are not statistically significant, as the confidence intervals for both North and South & Islands include 1.

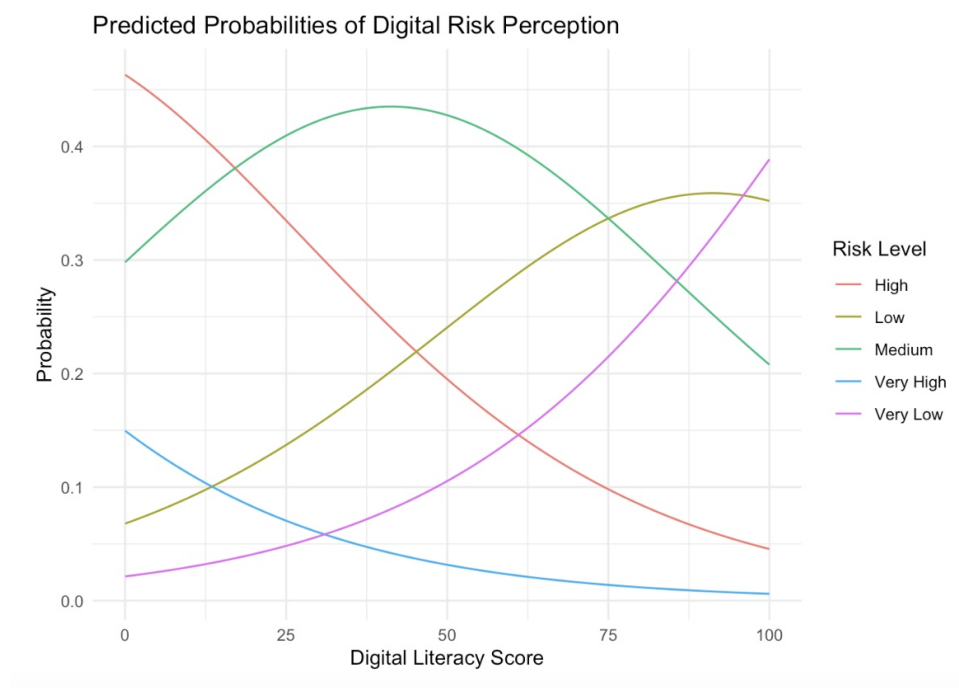


Figure 4: Predicted Probabilities of Digital Risk Perception

Figure Explanation Figure 4 illustrates the relationship between digital literacy scores and the predicted probabilities of different risk perception levels:

- As digital literacy scores increase (moving right on the x-axis), the probability of perceiving very high or high risk (red and orange lines) decreases sharply.
- Conversely, the probability of perceiving low or very low risk (light blue and dark blue lines) increases with higher digital literacy scores.
- The probability of perceiving medium risk (green line) shows a slight increase and then decrease, peaking at moderate levels of digital literacy.
- For individuals with very high digital literacy scores (right side of the graph), the most likely perception is low risk, followed by very low risk.
- For those with very low digital literacy scores (left side of the graph), high and very high risk perceptions are most probable.

This visualization clearly demonstrates the inverse relationship between digital financial literacy and perceived risk in digital financial contexts, supporting the regression results.

7 Conclusions and Recommendations

This analysis of digital financial literacy and risk perception in Italy yields important insights for education and policy-making. Based on the findings, I propose the following recommendations:

- **Tailored Educational Programs:** Develop digital financial literacy programs that account for the varying levels of literacy across different demographic groups. This research shows that education level is a strong predictor of digital financial literacy, suggesting that interventions should be tailored to different educational backgrounds. For instance, basic digital financial skills could be integrated into primary and secondary education curricula, while more advanced concepts could be introduced in higher education and adult learning programs.
- **Income-Based Interventions:** Given the positive relationship found between income and digital financial literacy, policymakers should focus on providing additional support and resources to lower-income groups. This could include subsidized access to digital financial tools and targeted educational programs in communities with lower average incomes.
- **Age-Specific Strategies:** Results also indicate that age is negatively associated with digital financial literacy but positively associated with risk perception. Educational initiatives should therefore be age-appropriate, with a focus on improving digital skills among older adults while also addressing overly cautious attitudes towards digital financial tools among this demographic.
- **Risk Education:** The inverse relationship between digital literacy and risk perception highlights the need for balanced risk education. As individuals become more digitally literate, it's crucial to ensure they maintain an appropriate level of caution. Educational programs should incorporate modules on identifying and managing digital financial risks, particularly for those with high digital literacy scores.
- **Gender-Sensitive Approaches:** While gender differences in the study were small, they were still significant. Policy makers and educators should consider these differences when designing interventions, ensuring that programs are equally accessible and appealing to all genders.
- **Regional Policy Adjustments:** Analyses also revealed regional disparities in digital financial literacy. Policymakers should consider region-specific interventions, potentially allocating more resources to areas with lower average digital financial literacy scores (southern regions).

7.1 Future research

Future research should focus on actually implementing and evaluating the effectiveness of the interventions by comparing the data of the successive years and continuing to

monitor the evolving landscape of financial and digital finance literacy in Italy, by finding new/more factors that might be impacting it.